

Lakshay Saini

+91-9718686175 | lakshaysaini210905@gmail.com | [linkedin.com/in/lakshay-saini/](https://www.linkedin.com/in/lakshay-saini/) github.com/LxSH9215

AI/ML ENGINEER

Innovative AI/ML Engineer specializing in Large Language Model (LLM) orchestration, Retrieval-Augmented Generation (RAG), Computer Vision, and Natural Language Processing. Demonstrated expertise in building production-grade AI systems, optimizing vector search pipelines, and deploying local AI models for automated analysis. Seeking to leverage strong problem-solving and prompt engineering skills to architect cutting-edge AI solutions.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++, Java, SQL

AI & Deep Learning: RAG Pipelines, LLM Orchestration, TensorFlow, PyTorch, Keras, CNNs, NLP, Computer Vision

Frameworks & Models: DeepSeek-Coder, Qwen, Ollama, Gemini API, OpenCV, PyQt5, Streamlit

Tools & Databases: ChromaDB (Vector Store), Docker, Git, REST API Development, Firebase, MySQL, Oracle

PROJECTS

Android Project Visualizer

2025 – Present

AI-Powered Architecture Analysis Tool

Python, Ollama, ChromaDB, Streamlit, RAG

- **Purpose:** Built to automate architectural discovery, reducing the steep learning curve, onboarding time, and technical debt for developers navigating complex, undocumented codebases.
- Architected a local RAG pipeline and multi-model LLM orchestration engine to reverse-engineer Android codebases into interactive architectural maps.
- Developed an end-to-end indexing pipeline with MD5 fingerprinting, parallel parsing, and vector embeddings, drastically reducing retrieval latency.
- Engineered a multi-layer LRU caching system and an intelligent query expansion module, maximizing retrieval recall while strictly enforcing LLM context budgets.
- Designed a production-grade HTTP client with connection pooling and token-streaming, integrating DeepSeek for security auditing and Qwen2.5 for structured UML generation.

Brahma: Intelligent Surveillance System

2024 – Present

Computer Vision Security Project

Python, OpenCV, TensorFlow

- **Purpose:** Developed to modernize physical security protocols, mitigating unauthorized access and significantly accelerating emergency response times.
- Engineered a real-time iris recognition system achieving 98% accuracy, reducing physical security breaches by 30%.
- Architected a highly scalable, GDPR-compliant data pipeline capable of handling 10,000+ biometric profiles.
- Integrated automated panic alerts, decreasing emergency response times to under 90 seconds.

Accentifi: Multilingual NLP Engine

2024

Natural Language Processing API

Python, Gemini API, Kotlin

- **Purpose:** Engineered to provide scalable, low-latency language translation while significantly cutting localization overhead costs for enterprise applications.
- Developed a language detection and translation system with 92% accuracy across 5 different languages.
- Created a low-latency API handling 50+ requests per second with sub-200ms response times.
- Implemented automated TTS generation, reducing translation operational costs by \$200/month.

SkinDoc: Dermatology AI

2023

Predictive Healthcare Modeling

TensorFlow Lite, Android Studio, CNN

- **Purpose:** Created to democratize preliminary dermatological screening and assist clinicians in reducing misdiagnosis rates.
- Trained a Convolutional Neural Network achieving 92% diagnostic accuracy on the industry-standard ISIC dataset.
- Reduced prescription errors by 45% through an automated diagnostic validation layer.
- Deployed the model to a mobile-friendly application, increasing partner clinic bookings by 30%.

EXPERIENCE

Core Member - Technical Department

Software Development Club, VIT Bhopal

Feb 2025 – July 2025

Bhopal, MP

- Mentored 15+ developers in Machine Learning best practices, advanced Python, and system design.
- Conducted code reviews and architectural planning, improving overall project code quality by 40%.

EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science (Specialization in AI & ML) | CGPA: 8.75/10

Bhopal, MP

Expected 2027

Jinvani Bharti Public School

Senior Secondary Education | 73.6%

New Delhi, Delhi

Graduated 2023

CERTIFICATIONS & ACHIEVEMENTS

- **National Semi-Finalist:** TATA Imagination Challenge 2025,24 (Ranked in the top 5% nationwide)
- **Certifications:** Advanced Python (GeeksforGeeks - Top 5%), Cloud Computing (NPTEL), Machine Learning (VIT Bhopal), AWS AI Fundamentals (In Progress)